Quantum Mechanics II — Exercise Sheet 5

Spectral readings and conditioning

Variant: Questions only

Exercise 1 (From momentum to energy density). A normalized line state has momentum density $(2b)^{-1}\mathbf{1}_{[-b,b]}$, where $b > 0$. (i) Find the cumulative distribution and density of its free energy $H_0 = P^2/(2m)$. (ii) Compute the median energy and the mean energy from this density.

Exercise 2 (A degenerate reading retains a phase). Let $f \in L^2(\mathbb{R})$ be normalized and supported in $(a, 2a)$ with $a > 0$. Set $g(x) = f(-x)$ and $\psi = (f + e^{i\phi}g)/\sqrt{2}$. Compare a reading of Q^2 in $(a^2, 4a^2)$ with a reading of the sign of Q whose result is forgotten. Give the resulting states and the expectation of parity $(J\chi)(x) = \chi(-x)$ in each.

Exercise 3 (A joint reading that is not independent). A normalized state has position density $1/(2a)$ on $(-a, a)$ and zero elsewhere. The commuting observables are Q and Q^2. (i) Find $\mathrm{Prob}(Q > 0, Q^2 < a^2/4)$ and compare it with the product of the marginal probabilities. (ii) Then replace $Q > 0$ by $Q > a/2$ and decide whether the two observables are independent in this state.

Exercise 4 (A finite first moment without a second). Take position density $w(x) = (1+|x|)^{-3}$ and the normalized vector $\psi = \sqrt{w}$. Check normalization, compute the mean position, and decide whether $\psi \in D(Q)$. Distinguish an absolutely convergent mean from a symmetric principal value.